

BRUTUS: Autonomous Neural Communication Intelligence System

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****Project Overview:****

BRUTUS is an advanced autonomous neural operating system built to bridge the gap between human intent and machine intelligence. BRUTUS transforms natural language into intelligent real-world digital actions while integrating powerful multilingual communication capabilities for seamless global interaction.

For this challenge, BRUTUS introduces a real-time multilingual AI communication framework that enables users to communicate across languages instantly through contextual speech and text translation.

****Problem Statement:****

Language barriers continue to hinder digital inclusion, education accessibility, and cross-border collaboration. Existing translation tools often suffer from:

- * Poor contextual understanding
- * Lack of cultural nuance preservation
- * High response latency
- * Weak support for regional/low-resource languages
- * Incompatibility with live meeting/chat platforms

****Solution:****

BRUTUS solves this by combining its autonomous neural intelligence architecture with an AI-powered multilingual communication engine featuring:

- * Real-Time Speech-to-Text & Text-to-Speech Pipeline: Enables instant voice input recognition and translated speech output.
- * Context-Aware Translation Engine: Uses LLM-driven semantic translation to maintain context, intent, and tone.
- * Support for Regional & Low-Resource Languages: Expandable multilingual model architecture designed for inclusivity.
- * Ultra Low-Latency Processing: Optimized inference pipeline ensures smooth live interactions.
- * Integration-Ready Architecture: Can integrate into meetings, messaging apps, education platforms, and enterprise communication systems.

****Key Innovation:****

What differentiates BRUTUS is its hybrid identity as both an Autonomous Neural OS and Communication Intelligence Platform, allowing it to not only translate conversations but also understand, reason, remember context, and execute intelligent workflows around communication.

?? Brutus AI: Features & Capabilities

Core Identity

Your AI assistant ? with a physical face. A fully-featured Windows desktop AI agent that controls a real humanoid robot head over Bluetooth.

****Tech Stack:**** Electron, React, Arduino, Groq, Windows License

****Capabilities:**** Voice conversations, Live lip-sync, Robot animations, OS automation, Web search, Email, Vision, Screen share, Deep research, File control, ADB mobile link, and more.

?? ****Also on Android?****

Brutus has a companion mobile app ? a full-featured Android AI assistant with its own robot BLE control, Gemini Live voice, and 25+ tools.

?? Project At A Glance

Metric Value
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??? AI Tools 40+ callable tools
?? Robot Animations 20 (10 macros + 10 tricks)
?? Expressions 6 (+ intensity slider 0?100%)
?? Servos 4 × SG90
?? AI Providers Groq (LLaMA 3), HuggingFace, Tavily, local Xenova
??? Voice Stack Whisper STT + Meta MMS / Kokoro TTS
??? Architecture Electron + React 19 + IPC bridge
?? Vector DB LanceDB
?? Mobile Link ADB over Wi-Fi

?? Architecture & Tech Stack

??? Architecture

```
`brutus-ai`  
??? `src`  
? ??? `main` (Electron Main Process: App entry, IPC, BLE, 40+ Tools, ADB, Gmail, LanceDB)  
? ??? `preload` (Context isolation)  
? ??? `renderer` (React 19 frontend, Tailwind v4, Zustand, Framer Motion, Three.js)
```

Tech Stack

Layer Technology
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Desktop runtime Electron 41.x

Frontend	React 19 + Tailwind CSS v4
Animations	Framer Motion + GSAP 3
LLM reasoning	Groq SDK (LLaMA 3 / Mixtral)
Local inference	@xenova/transformers
Vector DB	LanceDB
OS automation	Nut.js
Mobile	ADB over Wi-Fi

?? Hardware Robot

Brutus has a physical humanoid face using 4 micro servos, an LED, a sound sensor, and an HM-10 BLE module controlled by an Arduino Uno for real-time lip-sync and emotion driven by the desktop app.

?? Bill of Materials (Estimated Build Cost: ~\$15-25 USD)

- * Arduino Uno
- * HM-10 BLE Module
- * 4 × SG90 Micro Servos (Eyes, Eyelid, Mouth)
- * LED & Sound Sensor

?? BLE Protocol

Communication via BLE GATT serial (UUID 0000FFE1) with newline-terminated ASCII commands for Expressions (E), Mouth angle (M), Eye look-at (L), Blinks (B), Animations (A), and LED patterns (C).

?? Expressions (E command): Happy, Angry, Sad, Thinking, Sleepy, Surprised (Intensity 0-100%)

?? Animation Macros (A command) & Movement Tricks (W command): Nod, Shake, Wink, Laugh, Eye Roll, Jaw Drop, Crazy Eyes, and more.